

experimental techniques we cannot make the S_z — component out of the third apparatus in Figure 1.3c disappear. The peculiarities of quantum mechanics are imposed upon us by the experiment itself. The limitation is, in fact, inherent in microscopic phenomena.

1.1.3 Analogy with Polarization of Light

Because this situation looks so novel, some analogy with a familiar classical situation may be helpful here. To this end we now digress to consider the polarization of light waves. This analogy will help us develop a mathematical framework for formulating the postulates of quantum mechanics.

Consider a monochromatic light wave propagating in the z -direction. A linearly polarized (or plane polarized) light with a polarization vector in the x -direction, which we call for short an x -polarized light, has a space-time dependent electric field oscillating in the x -direction

$$\mathbf{E} = E_0 \hat{\mathbf{x}} \cos(kz - \omega t). \quad (1.5)$$

Likewise, we may consider a y -polarized light, also propagating in the z -direction,

$$\mathbf{E} = E_0 \hat{\mathbf{y}} \cos(kz - \omega t). \quad (1.6)$$

Polarized light beams of type (1.5) or (1.6) can be obtained by letting an unpolarized light beam go through a Polaroid filter. We call a filter that selects only beams polarized in the x -direction an x -filter. An x -filter, of course, becomes a y -filter when rotated by 90° about the propagation (z) direction. It is well known that when we let a light beam go through an x -filter and subsequently let it impinge on a y -filter, no light beam comes out provided, of course, we are dealing with 100% efficient Polaroids; see Figure 1.4a.

The situation is even more interesting if we insert between the x -filter and the y -filter yet another Polaroid that selects only a beam polarized in the direction — which we call the x' -direction — that makes an angle of 45° with the x -direction in the xy plane; see Figure 1.4b.

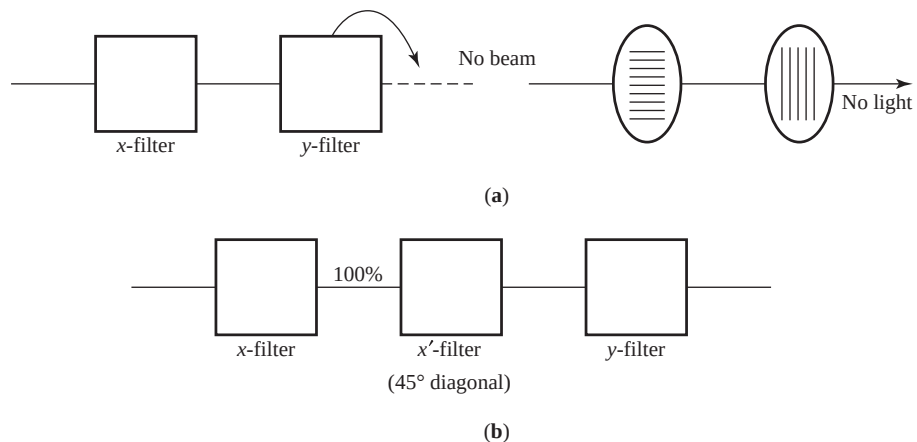


Fig. 1.4

Light beams subjected to Polaroid filters.

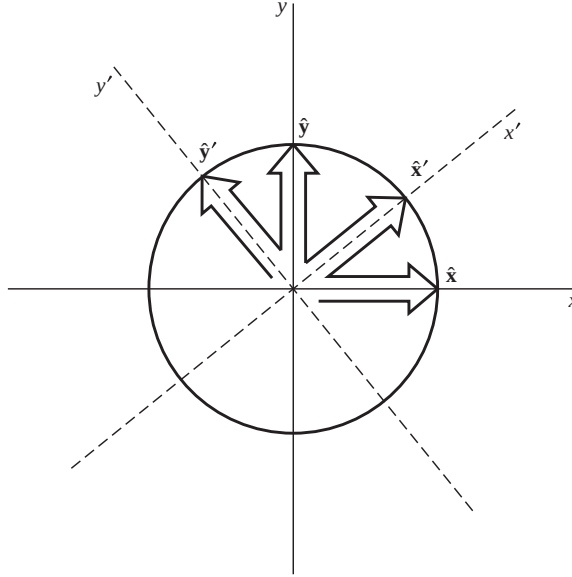


Fig. 1.5

Orientations of the x' - and y' -axes.

This time, there is a light beam coming out of the y -filter despite the fact that right after the beam went through the x -filter it did not have any polarization component in the y -direction. In other words, once the x' -filter intervenes and selects the x' -polarized beam, it is immaterial whether the beam was previously x -polarized. The selection of the x' -polarized beam by the second Polaroid destroys any previous information on light polarization. Notice that this situation is quite analogous to the situation that we encountered earlier with the SG arrangement of Figure 1.3b, provided that the following correspondence is made:

$$\begin{aligned} S_z \pm \text{atoms} &\leftrightarrow x\text{-, } y\text{-polarized light} \\ S_x \pm \text{atoms} &\leftrightarrow x'\text{-, } y'\text{-polarized light,} \end{aligned} \quad (1.7)$$

where the x' - and the y' -axes are defined as in Figure 1.5.

Let us examine how we can quantitatively describe the behavior of 45° -polarized beams (x' - and y' -polarized beams) within the framework of classical electrodynamics. Using Figure 1.5 we obtain

$$\begin{aligned} E_0 \hat{\mathbf{x}}' \cos(kz - \omega t) &= E_0 \left[\frac{1}{\sqrt{2}} \hat{\mathbf{x}} \cos(kz - \omega t) + \frac{1}{\sqrt{2}} \hat{\mathbf{y}} \cos(kz - \omega t) \right], \\ E_0 \hat{\mathbf{y}}' \cos(kz - \omega t) &= E_0 \left[-\frac{1}{\sqrt{2}} \hat{\mathbf{x}} \cos(kz - \omega t) + \frac{1}{\sqrt{2}} \hat{\mathbf{y}} \cos(kz - \omega t) \right]. \end{aligned} \quad (1.8)$$

In the triple-filter arrangement of Figure 1.4b the beam coming out of the first Polaroid is an $\hat{\mathbf{x}}$ -polarized beam, which can be regarded as a linear combination of an x' -polarized beam and a y' -polarized beam. The second Polaroid selects the x' -polarized beam, which can in turn be regarded as a linear combination of an x -polarized and a y -polarized beam. And finally, the third Polaroid selects the y -polarized component.

Applying correspondence (1.7) from the sequential Stern–Gerlach experiment of Figure 1.3c, to the triple-filter experiment of Figure 1.4b suggests that we might be able to represent the spin state of a silver atom by some kind of vector in a new kind of two-dimensional vector space, an abstract vector space not to be confused with the usual two-dimensional (xy) space. Just as $\hat{x}$ and $\hat{y}$ in (1.8) are the base vectors used to decompose the polarization vector $\hat{x}'$ of the $\hat{x}'$ -polarized light, it is reasonable to represent the S_x+ state by a vector, which we call a *ket* in the Dirac notation to be developed fully in the next section. We denote this vector by $|S_x; +\rangle$ and write it as a linear combination of two base vectors, $|S_z; +\rangle$ and $|S_z; -\rangle$, which correspond to the S_z+ and the S_z- states, respectively. So we may conjecture

$$|S_x; +\rangle \stackrel{?}{=} \frac{1}{\sqrt{2}}|S_z; +\rangle + \frac{1}{\sqrt{2}}|S_z; -\rangle \quad (1.9a)$$

$$|S_x; -\rangle \stackrel{?}{=} -\frac{1}{\sqrt{2}}|S_z; +\rangle + \frac{1}{\sqrt{2}}|S_z; -\rangle \quad (1.9b)$$

in analogy with (1.8). Later we will show how to obtain these expressions using the general formalism of quantum mechanics.

Thus the unblocked component coming out of the second (SG $\hat{x}$) apparatus of Figure 1.3c is to be regarded as a superposition of S_z+ and S_z- in the sense of (1.9a). It is for this reason that two components emerge from the third (SG $\hat{z}$) apparatus.

The next question of immediate concern is: How are we going to represent the $S_y\pm$ states? Symmetry arguments suggest that if we observe an $S_z\pm$ beam going in the x -direction and subject it to an SG $\hat{y}$ apparatus, the resulting situation will be very similar to the case where an $S_z\pm$ beam going in the y -direction is subjected to an SG $\hat{x}$ apparatus. The kets for $S_y\pm$ should then be regarded as a linear combination of $|S_z; \pm\rangle$, but it appears from (1.9) that we have already used up the available possibilities in writing $|S_x; \pm\rangle$. How can our vector space formalism distinguish $S_y\pm$ states from $S_x\pm$ states?

An analogy with polarized light again rescues us here. This time we consider a circularly polarized beam of light, which can be obtained by letting a linearly polarized light pass through a quarter-wave plate. When we pass such a circularly polarized light through an x -filter or a y -filter, we again obtain either an x -polarized beam or a y -polarized beam of equal intensity. Yet everybody knows that the circularly polarized light is totally different from the 45°-linearly polarized (x' -polarized or y' -polarized) light.

Mathematically, how do we represent a circularly polarized light? A right circularly polarized light is nothing more than a linear combination of an x -polarized light and a y -polarized light, where the oscillation of the electric field for the y -polarized component is 90° out of phase with that of the x -polarized component:⁴

$$\mathbf{E} = E_0 \left[\frac{1}{\sqrt{2}} \hat{x} \cos(kz - \omega t) + \frac{1}{\sqrt{2}} \hat{y} \cos\left(kz - \omega t + \frac{\pi}{2}\right) \right]. \quad (1.10)$$

⁴ Unfortunately, there is no unanimity in the definition of right versus left circularly polarized light in the literature.